**CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT**

Course Code and Title**:**S-CSGD415LA Advanced Game Programming Lab

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

**DIRECTIONS:** Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

| Format, parts, and description | Compliance | |
|---|---|---|
| | **Complied With** | **Not Complied With** |
| **Format** | | |
| 1. The 2023-2024 syllabus format is used. | / | |
| 2. The font used in the syllabus preparation is Times New Roman (11). | / | |
| 3. DLSUD (being first) and college logos are beside each other at the top (center). | / | |
| 4. Preliminaries include all the contact details of the professor, such as the day and time. | / | |
| 5. Each WEEK has a corresponding date. | / | |
| **Content** | | |
| 6. The Learning Outcomes have more HOTS than LOTS. | / | |
| 7. CLOs and TLOs are identified for each module/lesson/topic. | / | |
| 8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities. | / | |
| 9. The journal articles/references are within the last five years. | / | |
| 10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies. | / | |
| 11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students. | / | |

| | | |
|---|---|---|
| **Assessments** | | |
| 12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus. | / | |
| 13. Formative Assessments are not reflected anymore in the syllabus. | / | |
| 14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities. | / | |
| 15. The syllabus presents one-module(lesson)-one-assessment correspondence. | / | |
| 16. Only one Enabling Assessment is reflected in a given week | / | |
| 17. The Enabling Assessments are aligned with the identified CLOs and TLOs. | / | |
| 18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term. | / | |
| 19. Thirty percent (30%) is allotted for Summative Assessments per term. | / | |
| 20. Fifteen percent 20%) is allotted for Class Participation per term. | / | |
| 21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade. | / | |
| 22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities. | / | |
| 23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities. | / | |
| 24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD | / | |
| 25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD. | / | |
| **References and Curriculum Map** | | |
| 26. The References must be within the last 5 years except for canons and seminal studies. | / | |
| 27. It must be a combination of books, eBooks, journal articles, videos, etc. | / | |
| 28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS. | / | |
| 29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs) | / | |

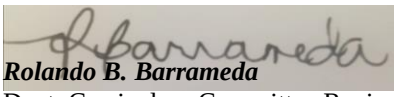
NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

**Self-reviewed:**



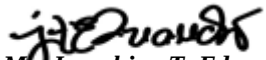
***Ms. Josephine T. Eduardo***
Faculty

**Checked, verified, and endorsed:**



***Rolando B. Barrameda***
Dept. Curriculum Committee Reviewer

**Checked, verified, and approved:**



***Ms. Josephine T. Eduardo***
Department Chair



***Dr. Rubie M. Causaren***
College Dean

**DE LA SALLE UNIVERSITY-DASMARINAS**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER STUDIES DEPARTMENT**

## COURSE SYLLABUS

| | | |
|---|---|---|
| **COURSE CODE** | **:** | S-CSGD415LA |
| **COURSE TITLE** | **:** | Advanced Game Programming |
| **COURSE TYPE** | **:** | Laboratory |
| **COURSE CREDIT** | **:** | 1 Unit |
| **MODALITY** | **:** | **Fully Onsite** |
| **PRE-REQUISITES** | **:** | S-CSGD323LA |
| **CO-REQUISITES** | **:** | None |
| **PROFESSOR** | **:** | **MS. JOSEPHINE T. EDUARDO** |
| **CONSULTATION** | **:** | **H/1300-1500/COS100-C/jteduardo@dlsud.edu.ph/ MS Teams/Schoolbook** |

**COURSE DESCRIPTION:**

This course deals with advanced game programming with a strong focus on developing good programming skills, design patterns, and practical problem-solving.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on the national, regional, provincial, city, and institutional guidelines for the safety amidst disaster and pandemic. Hence, this class shall be delivered in both online and offline/module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

**COURSE LEARNING OUTCOMES:**

By the end of this course, students can:

CLO1. understand the structure of mobile game programming;
CLO2. remember the basic framework for mobile game development;
CLO3. understand how physics affects in development of the game;
CLO4. apply the design learned and report the expected effectiveness of the mobile game;
CLO5. demonstrate game programming capability;
CLO6. interpret efficient game loop;
CLO7. evaluate mobile game application;
CLO8. design the needs of game programming in the industry;
CLO9. develop an appreciation in virtual simulation technology; and
CLO10. value the Lasallian values and sensitivity in developing a game to real-world problems.

**LEARNING PLAN**

**Learning Outcomes for Midterm Period**

| **Course Learning Outcomes** | **Topic Learning Outcomes** |
|---|---|
| CLO1. Understand the structure of mobile game programming. | TLO1. Explain basic mobile game in a nutshell. |
| CLO2. Remember the basic framework for mobile game development. | TLO2. Apply key concepts of game processes, principles and ethics. |
| CLO3. Understand how physics affects in development of the game. | TLO3. Apply working with frameworks. |
| CLO 4. Apply the design learned and report the expected effectiveness of the mobile game. | TLO4. Detemine and generate mobile game programming. |
| CLO5. Demonstrate game programming capability. | TLO5. Support observation to mobile game programming. |
| CLO6. Interpret efficient game loop. | TLO6. Classify the domain areas of mobile game programming. |

| **Module and Learning Outcomes** | **Week** | **Teaching-Learning Activities / Assessment Strategies** | **Technology Enabler** | **Onsite** | **Allotted time** |
|---|---|---|---|---|---|
| Introduction to Module | Week 1<br>Aug 14-19 | Gospel Reading<br>Course Syllabus<br>Course Overview<br>Enabling Assessment:<br>• Introduction to yourself<br>• Course Expectation<br>Laboratory Rules and Familiarization with the software | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 1<br>CLO1, CLO2, TLO1, TLO2 | Week 2<br>Aug 21-26 | Gospel Reading<br>Module 1:<br>• Unity and C#<br>• Why mobile?<br>• Unity & C#: Game Dev Basics, Scenes, Game Objects, and Components<br>Enabling Assessment<br>• Laboratory Activity 1 Installation of Framework | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 2<br>CLO2, CLO3, TLO2, TLO3 | Week 3<br>Aug 28-<br>Sept 2 | Gospel Reading<br>Module 2:<br>• Installation of Unity and Android<br>Enabling Assessment<br>• Laboratory Activity 2 Familiarization with the Framework | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 3<br>CLO3, CLO4 | Week 4<br>Sept 4- | Gospel Reading<br>Module 3: | Schoolbook<br>Powerpoint | / | 2.0 |

| | | | | | |
|---|---|---|---|---|---|
| TLO3, TLO4 | Sept 9 | • Camera<br>• Sprites<br>• Texture Assets and Atlases<br>• Animation<br>• Scrolling<br>Enabling Assessment<br>• Laboratory Activity 3 on Module 3 | MS Teams | | |
| Module 4<br>CLO2, CLO3, CLO4<br>TLO2, TLO3, TLO4 | Week 5<br>Sept 11 – Sept 16 | Gospel Reading<br>Module 4:<br>• Math, and 3D Introduction<br>• Overview of Vector Math<br>• 3D Math Primer<br>• Basics of the 3D World<br>• 3D Rendering Essentials<br>• Unity for 3D Development<br>Enabling Assessment<br>• Laboratory Activity 4 on Module 4 | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 5<br>CLO4, CLO5<br>TLO4, TLO5 | Week 6<br>Sept 18 – Sept 23 | Gospel Reading<br>Tasks: Reading on Module 5<br>• Mobile Game Input; Designing for Mobile<br>• Basic Touch and Multi-task Gestures<br>• Accelerometer<br>• Virtual Joy Pads<br>Enabling Assessment<br>• Laboratory Activity 5 on Module 5 | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 6<br>CLO5, CLO6<br>TLO5, TLO6 | Week 7<br>Sept 25 – Sept 30 | Gospel Reading<br>Module 6:<br>• Audio<br>• Basic Physics Principle<br>Enabling Assessment<br>• Laboratory Activity 6 on Module 6 | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| **WEEK 8 (Oct 9 – 14)**<br>**MIDTERM EXAMINATIONS** | | | | | |
| | | | **TOTAL** | **18** | **18.0** |

| Course Learning Outcomes | Topic Learning Outcomes |
|---|---|
| CLO7. Evaluate mobile game application. | TLO7. Exhibit and demonstrate mobile game programming. |
| CLO8. Design the needs of game programming in the industry. | TLO8. Develop games in Android mobile platform. |
| CLO9. Develop an appreciation in virtual simulation technology. | TLO9. Collaboration, individual assessment to design and develop group project. |
| CLO10. Value the Lasallian values and sensitivity in developing a game to real-world problems. | TLO10. Collaboration and presentation of group project. |

| Module and Learning Outcomes | Week | Teaching-Learning Activities / Assessment Strategies | Technology Enabler | Onsite | Allotted time |
|---|---|---|---|---|---|
| **Week 10 (Oct 16 – 21)<br>STUDENT'S WEEK** | | | | | |
| **Week 11 (Oct 23 – 28)<br>SELF-CARE WEEK** | | | | | |
| Module 7 | Week 12<br>Oct 30 – Nov 4 | Gospel Reading<br>Module 8:<br>• Particles Effect<br>• Cross-Platform Game Engines<br>• Platform Specific Game Creation<br><br>Tasks: Work on Team Project and Progress Report | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 8 | Week 13<br>Nov 6 - 11 | Gospel Reading<br>Module 9:<br>• Tile Maps<br>• AI Behavior<br>• Path Finding<br>Tasks: Work on Team Project and Progress Report<br>Enabling Assessment: Progress Report on Game Development | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 9 | Week 14<br>Nov 13 -18 | Gospel Reading<br>Module 10:<br>• Augmented and Virtual Reality Games Design and Principles<br>Enabling Assessment: Progress Report on Game Development<br>Tasks: Work on Team Project and Progress Report | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 10 | Week 15<br>Nov 20 - 25 | Gospel Reading<br>Module 11:<br>• Shaders on Mobile | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |

| | | | | | |
|---|---|---|---|---|---|
| | | • Advanced 3D Effects<br>• Plugins<br>• Deploying on the App Store<br>Tasks: Work on Team Project and Progress Report | | | |
| Module 11 | Week 16<br>Nov 27 – Dec 2 | Gospel Reading<br>Module 12<br>Tasks: Work on Team Project and Progress Report | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| Module 12 | Week 17<br>Dec 4 – 9 | Gospel Reading<br>Group Project Presentation | Schoolbook<br>Powerpoint<br>MS Teams | / | 2.0 |
| **Week 18 (Dec 11 – 16)<br>FINAL EXAM** | | | | | |
| | | | **TOTAL** | **18** | **18.0** |

**GRADING SYSTEM;**

**Midterm**

| | |
|---|---|
| Enabling Assessments | - 50% |
| Attendance/Class Participation | - 20% |
| Summative Assessments | - 30% |
| | **100%** |

**Finals**

| | |
|---|---|
| Enabling Assessments | - 50% |
| Attendance/Class Participation | - 20% |
| Summative Assessments | - 30% |
| | **100%** |

Overall Final Grade = Midterm + Finals / 2

**COURSE POLICIES AND REQUIREMENTS;**

1. **Office365 Activation.** Please ensure that your Office365 account is working. Your Office365 account is needed to access both Schoolbook and MS Teams where your asynchronous and synchronous classes will be held.

2. **Enrollment in an E-Class.** You will automatically be enrolled in your e-class which is based on your enrollment data.

3. **Enriched Virtual Blended Learning Model.** This course adopts the enriched virtual blended model. This means that there will be a mix of face-to-face, synchronous, and asynchronous activities. Majority of teaching-learning activity and assessments are undertaken online. The total number of onsite classes shall be 30% of the number hours allotted for the whole semester.

4. **Online Asynchronous Sessions.**
    a. **Schoolbook (SB).** Schoolbook shall be the only platform for asynchronous sessions.

b. **Modules.** Modules are self-paced learning resources for asynchronous sessions. These can be accessed in Schoolbook.
c. **References.** Each page section may contain uploaded references. These learning resources may be downloaded.
d. **Asynchronous Activities.** You are expected to read the modules as soon as they are uploaded. The learning content of the modules complement the online synchronous and face-to-face sessions.
e. **Asynchronous Engagement.** Your activities in the course can be tracked by your professor. This includes the time you spend in reading the lessons and answering the assessments.
f. **Schoolbook Forum.** All general concerns about the lessons and assessments in asynchronous sessions must be posted in the Schoolbook Forum. Response shall be made by your teacher within 48 hours.
g. **Schoolbook Messaging.** This shall be the mode of communication for private and/or confidential communications. Response shall be made by your teacher within 48 hours upon receipt of the same unless it falls on weekends or holidays, which shall be handled promptly the following working day.

5. **Online Synchronous Sessions.**

a. **MS Teams.** MS Teams shall be the only platform for all online synchronous sessions.
b. **Gospel Reading and Prayer.** Each online synchronous session shall start with a Gospel reading and prayer. Your teacher may assign you, in advance, to do this.
c. **Synchronous Meeting Schedule.** The meeting schedule shall follow the time indicated in your official registration. The dates of synchronous meetings are identified in the learning plan. A scheduled synchronous class shall push through regardless of the number of attendees.
d. **Non-use of Official DLSU-D Office 365 account.** All students are required to join the MS Teams session using their official Office 365 account. Guest accounts will not be allowed to join the meeting.
e. **Excuse from online synchronous classes.** Students may be excused due to valid reasons such as disruptions of internet connections, serious illness, death in the family, and any other reasons analogous to the foregoing. Students will only be excused upon submission of proof of excuse before the next synchronous OR face-to-face meeting.
f. **Attendance.** Attendance at online synchronous meetings is required. Students shall be excused from attending synchronous classes if he/she has face-to-face classes on the same day.
g. **Tardiness.** A student who came 1 minute to 15 minutes after the start of the synchronous meeting is considered late. Three tardy attendance is equivalent to one absence.
h. **Absence.** A student is considered absent if the log-in time is beyond the class's first fifteen minutes. 40% absence in the entire semester means a 0.0 grade in this course. The duration of students' attendance in the meeting can be verified through the Insights feature of MS Teams.
i. **Review-Feedback-Preview.** Synchronous sessions shall utilize the Review-Feedback-Preview Model. Synchronous sessions are spent for discussion of difficult topics in the uploaded modules, a review of important concepts in the previous modules, feedback of students on the concepts learned, feedback of the teachers on the activities submitted by the students, and a preview of the succeeding modules/topics to be uploaded.
j. **Turning on of camera.** Turning on of camera is required during synchronous meetings, specifically at the start, end, and during interactions between students and teacher.

k. **Attire.** Students should wear attire as prescribed by the Student Manual.

l. **Recording of synchronous meetings.** Upon agreement among the students, all synchronous classes shall be recorded by the teacher. Only the teacher shall record MS Teams meetings. Students may access the recording at the MS Teams.

m. **Office 365 Photo.** Modify the settings of your Office365 account and upload your photo so that during MS Teams meeting, even if your camera is turned off, your teacher can recognize you.

n. **Microphone use.** Microphone must be muted and only unmuted when permitted by the teacher.

6. **Onsite / Face-to-face (F2F) Sessions.**

a. **Face-to-face engagement.** Your engagement in face-to-face classes is graded based on your class participation.
b. **Classroom.** F2F classes shall be held at the classroom indicated in your Certificate of Registration. Should there be changes in the classroom venue, information will be given in advance.
c. **Gospel Reading and Prayer.** Each F2F session shall start with a Gospel reading and prayer. Your teacher may assign you, in advance, to do this.
d. **F2F Meeting Schedule.** The meeting schedule shall follow the time indicated in your official registration. The dates of F2F meetings are identified in the learning plan.
e. **Attendance.** Attendance in F2F meetings is required.
f. **Tardiness.** A student who came 1 minute to 15 minutes after the start of the face-to-face meeting is considered late. Three tardy attendance is equivalent to 1 absence.
g. **Absence.** A student is considered absent 15 minutes after the official class schedule.
h. **Excuse from F2F classes.** Students are excused in the F2F classes based on the provisions in the latest version of the Student Handbook.
i. **Uniform.** Wearing of uniform is optional. Civilian attire should follow the policy on dress code as stipulated in the latest version of the Student Handbook.

7. **Assessment and Grading System.**

a. **Formative assessments.** These are ungraded assessments. These may be considered as practice assessments that leads towards achieving outcomes without fear of receiving a failing grade.
b. **Enabling assessments.** These will comprise most of your graded assessments. These are designed to achieve topic learning outcomes, that leads towards achieving the course learning outcomes. Only one enabling assessment shall be allowed during the week. Please pay attention to the duration and number of attempts. As a general rule, quiz-type enabling assessments shall be open for only a minimum of 24 hours, while output-based enabling assessments shall be open for at least 6 days.
c. **No. of Attempts.** All enabling assessments, whether given onsite or online, shall have a

maximum of 3 attempts only. Summative assessments shall be given a maximum of 2 attempts.

d. **Summative assessments.**
    i. There shall be two summative assessments (midterm and final exams) for the entire semester. These are designed to achieve the course learning outcomes.
    ii. Online summative assessments shall be uploaded on SB and shall be accessible for a minimum of 1 day for quiz type and a minimum of 6 days for output-based assessments.
    iii. Summative assessment maybe given through online or face-to-face.
    iv. Output-based summative assessment shall be given to students at least fifteen days prior to scheduled Summative Exam Week.
e. **Lifeline.** Only students with (1) valid reason as stated in the Student Handbook and IRR, and (2) given their proof of excuse on or before the next synchronous/F2F session, shall be given a lifeline on the enabling and summative assessments.
f. **Rubric.** All online non-quiz or non-discrete types of assessment (essay, drop box, output-based, etc.) shall have a rubric or criteria for rating the students' tasks. A student may refuse to answer these types of assessment in the absence of a rubric or criteria for grading, and the assessment shall be deemed invalid and shall not be part of the student's grades.

g. **Grading.** All online assessments should be checked and graded by the teacher before the submission of midterm and final grades.

h. **Grading system.** The following shall be the basis for the computation of grades per term for fully online and enriched virtual blended classes (see table above):

   i. Enabling Assessments: 50%

   ii. Class Participation (including Attendance): 20%

   iii. Summative Assessments: 30%

i. **Gradebook.** Students can see the breakdown of grades in their Assessment tab.

8. **Self-Care.**

a. **Schedule.** There are two schedules of Self-Care week during the First Semester SY2022-2023. The first is on Sept. 26 to Oct. 1, and the second is on Nov. 28 to Dec. 3. During these weeks, there shall be no synchronous meetings, F2F classes, new modules, new assessments, and deadlines.

b. **Prerogative.** Students may avail of the self-care program, whether online or onsite, provided by the different units of the University.

9. **Data Privacy.**

a. **On camera.** Your teacher may require you to engage in synchronous session by turning on your camera. These type of engagement activities include, but not limited to the following: chat participation, participating by audio/voice, participation in poll and games, clicking links, and other online engagement activities.

b. **Recording of synchronous session.** Your teacher shall inform you, at the start of the meeting, that the synchronous meeting is being recorded.

c. **Access to the MS Teams recording.** Only students who are officially enrolled shall be part of the MS Teams and have access to all the resources including the recording. Students are not allowed to record the meetings. Students are not allowed to download the recordings. Screen recording is not allowed.

d. **Guests.** Inviting people, that are not part of the class, in synchronous meetings is strictly prohibited, unless approved by the subject teacher.

e. Posing as another person during a synchronous activity is strictly prohibited.

10. **Copyright and Plagiarism.**

a. Using of any illegally obtained software and other technology tool is strictly prohibited.

b. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.

c. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

d. This subject shall abide by the policies pertaining to intellectual property, copyright, and plagiarism as stipulated in latest edition of the Student Handbook.

e. Any plagiarized work, whether in part or full, shall mean a grade of 0.0 for the assessment.

11. This course shall abide by any institutional policies that may be released after the approval of this syllabus. Any such policy shall be posted within the e-class at the forums section, news feed. It will also be briefly discussed during the soonest synchronous meeting.

**REFERENCES;**

**Online References:**

| E- Provider | References |
|---|---|
| https://developer.android.com › games | Android Game Development |
| https://unity.com/solutions/mobile | Mobile Game Development Software - Unity |
| https://gdevelop-app.com/ | G-Develop Create Games without Programming Opensource |
| https://dl.icdst.org/pdfs/files3/83688bd08d219d5b85665fd41c060f05.pdf<br>https://learn.unity.com/tutorials | |

**Journal References**

Tecson P, Herrera A, Eduardo JT, Rosas MF, Barrameda RB, "Utilizing Simultaneous Localization and Mapping (SLAM) in Augmented Reality Shell Game for IOS" 2019 Test Engineering and Management
http://www.testmagzine.biz/index.php/testmagzine/article/view/696

Lalas GL, Canto KA, Uy GCN, Eduardo JT, Rosas MF, "Save Your Grades! A 3D Mobile Educational-Entertainment Game Using Ant Colony Algorithm" 2019 Test Engineering and Management
http://www.testmagzine.biz/index.php/testmagzine/article/view/728

**Prepared:** **Computer Studies Department**
1st Semester 2022-2023

**Approved:** **Josephine Eduardo**
Chair, CSD

**Endorsed:** **Dr. Ruble M. Causaren**
Dean, CSCS

**DE LA SALLE UNIVERSITY-DASMARINAS**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME**
**MAPPING TABLE FOR BS IN COMPUTER SCIENCE**

| | |
|---|---|
| **COURSE CODE:** | **S-CSGD415LA** |
| **COURSE TITLE:** | **ADVANCED GAME PROGRAMMING** |
| **COURSE TYPE:** | **LABORATORY** |
| **COURSE CREDIT:** | **3 UNITS** |
| **PRE-REQUISITES:** | **S-CSGD323** |
| **CO-REQUISITES:** | **NONE** |

| **COURSE LEARNING OUTCOMES** | **PROGRAM LEARNING OUTCOMES** | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | **PLO1** | **PLO2** | **PLO3** | **PLO4** | **PLO5** | **PLO6** | **PLO7** | **PLO8** |
| CLO1. Understand the structure of mobile game programming. | ✓ | | | ✓ | | | | |
| CLO2. Remember the basic framework for mobile game development. | ✓ | | ✓ | | | | | |
| CLO3. Understand how physics affects in development of the game | | | | | | ✓ | | |
| CLO 4. Apply the design learned and report the expected effectiveness of the mobile game. | | ✓ | | | ✓ | | | |
| CLO5. Demonstrate game programming capability. | ✓ | ✓ | ✓ | | | | | |
| CLO6. Interpret efficient game loop. | | ✓ | ✓ | | | | | |
| CLO7. Evaluate mobile game application. | | | | | | | | ✓ |
| CLO8. Design the needs of game programming in the industry. | | | ✓ | | ✓ | | | |
| CLO9. Develop an appreciation in virtual simulation technology. | | ✓ | | | ✓ | | | ✓ |
| CLO10. Value the Lasallian values and sensitivity in developing a game to real-world problems. | | | ✓ | | ✓ | ✓ | | ✓ |

*NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)*

**DE LA SALLE UNIVERSITY-DASMARINAS**
**COLLEGE OF SCIENCE AND COMPUTER STUDIES**
**COMPUTER SCIENCE DEPARTMENT**

## GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

| Program Learning Outcomes | Graduate Attributes/Institutional Learning Outcomes | | | | |
|---|---|---|---|---|---|
| | GA1 | GA2 | GA3 | GA4 | GA5 |
| 1. Apply concepts of computing and mathematics in different domains of Computer Science. | | | | ✓ | ✓ |
| 2. Identify and analyze problems and define computing requirements appropriate to their solutions. | | | | | ✓ |
| 3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources. | | | ✓ | ✓ | ✓ |
| 4. Document computer-based systems by writing effective reports for risk-assessment impact | | | | ✓ | ✓ |
| 5. Work effectively as an individual or as a member of a team during software development. | | | ✓ | ✓ | ✓ |
| 6. Communicate effectively with the computing community and non-computer science professionals about computing activities. | | | | ✓ | ✓ |
| 7. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues. | ✓ | ✓ | ✓ | | |
| 8. Engage in continuing professional development through performing independent or collaborative research as computing professionals. | | | ✓ | | |

*NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.*

**GRADUATES ATTRIBUTES AND INSTITUTIONAL LEARNING OUTCOMES (ILOs)**

| **Graduate Attribute (GA)** | **Descriptors (Institutional Learning Outcome)[1]** |
|---|---|
| 1. God-centered | a) Creates an environment where the experience of God is lived and shared<br>b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives)<br>c) Observes and maintains ethical standards in dealing with the different stakeholders<br>d) Integrates Christian perspectives and values in all undertakings<br>e) Manifests humility and respect in relating with other people |
| 2. Patriotic | a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives<br>b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts<br>c) Engages actively in political, social, economic, and cultural transformation for nation building<br>d) Brings pride and honor to the community and the country<br>e) Patronizes locally produced products and promotes them globally |
| 3. Socially Responsive | a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation<br>b) Engages in ecological advocacies as responsible stewards of God's creation<br>c) Continuously works in solidarity with people and institutions to effect liberating action<br>d) Brings forth awareness and deeper understanding of social realities<br>e) Manifests spirit of volunteerism through social actions that improve communities |
| 4. Globally Competitive | a) Takes progressive responsibility for own learning and development<br>b) Promotes critical and creative thinking, self-knowledge and self-mastery<br>c) Is committed to rise above mediocrity<br>d) Shows dedication, commitment and love for work to attain exceptional results<br>e) Exhibits innovativeness and creativity in various contexts |
| 5.Professionally Competent | a) Possesses leadership qualities of Competence, Honesty, Accountability, Responsibility and Transparency (CHART).<br>b) Effectively communicates in various forms of media<br>c) Strives to be at the top of their chosen fields<br>d) Effectively works and collaborates in a multi-disciplinary team<br>e) Adapts to changes in the workplace, local society and global community |

[1] *Approved in July 2017 during a multi-sectoral committee specifically convened for the purpose of coming up with descriptions for the graduate attributes.*